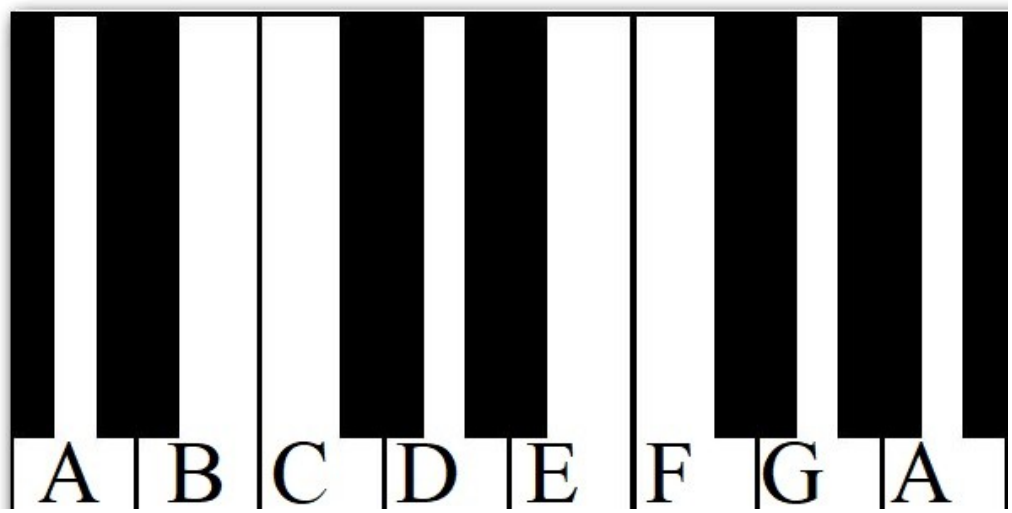
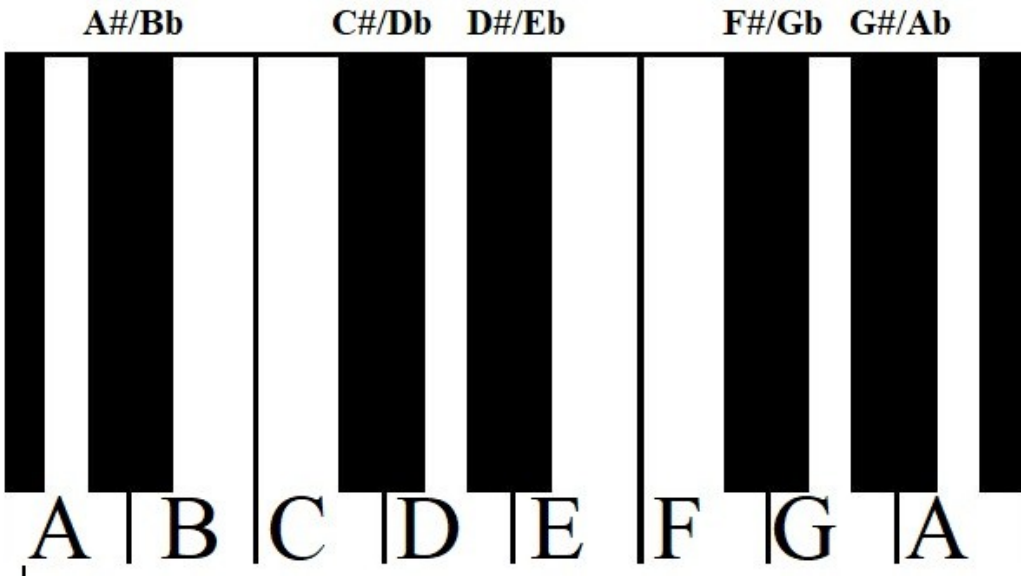
The Basics of Music

When thinking of music, we think of **notes**. Each instrument – and the human voice – produces sound in different ways, but all of them work through vibrations in the air that reach our ears. As students of theory, we identify notes by assigning them names like A B C D E F G, Do Re Mi, Sa Re Ga etc. Here, we will be using the English alphabet up to the letter G to signify the 7 Natural notes of music. Natural notes represent the white keys on a keyboard.



How do we find the black notes on a piano? We do so using a concept known as accidentals.

Accidentals are *symbols* attached to notes to signify movement
# = sharp = moving forward *(e.g. C# means a note ahead from C)*
b = flat = moving back *(e.g. Bb means a note behind B)*
♮ = natural = as is *(e.g. C, D, E....)*



Chromatic Scale = The scale that has all 12 notes present in Western Music Theory
e.g. A A# B C C#..... or A Bb B C Db.....

Knowing the notes that are used in music, we can now focus on grouping notes together and creating scales, or chords. But first, we have to learn the distances between the notes.

Intervals refer to the *gaps* or *spaces* between the notes.
The smallest intervals are:
1. Semitone/Half-step (S/H) *(e.g. C to C#)*
2. Tone/Whole-Step (T/W) *(e.g. C to D)*
[The Chromatic Scale is made using only semitones – E F F# G G# A....

An important concept to remember is called:
Enharmonic Equivalence = same sound / different name
e.g. E# is the same note as F, C# is the same note as Db

<u>Major and Minor Scales</u>

We can make any scale using a *formula* which is written in *intervals.*

The **Major Scale formula**, therefore is:
TTSTTTS / WWHWWWH

You can <u>start on any note and apply the intervals one by one</u> to see how we make a major scale.

<u>e.g.</u>	**C**	**D**	**E**	**F**	**G**	**A**	**B**	**C**	**=** C major scale
	T	*T*	*S*	*T*	*T*	*T*	*S*		
	1	2	3	4	5	6	7	8	

Once we've made a scale, we can number the notes to make it easier to change to different scales. When we number the notes, those numbers are called *degrees.*

*[It is important to note that all of our major and minor scales (as well as modes) go in **alphabetical order.** This means we cannot skip notes (A C D E) or repeat them (F G A A#). The scale also has to begin and end with the same note]*

Other scales (and modes) have their own formula.

For example, the **Minor Scale formula** would be:
TSTTSTT / WHWWHWW

With this, we should understand the concept of a **Relative Minor.** As the name suggests, it is a minor scale related to a major scale. This means that *a major scale will share the same notes as its relative minor.*

The Relative Minor is found at the 6th degree of the major scale. Conversely, the "relative major" is found at the 3rd degree of the minor scales due to the rule of *inversions.*

<u>e.g.</u> The 6th note of the C major scale is A. If we start on that note and continue using the notes of the C major scale until we reach A again, we will have an *A Natural Minor scale. A to C is a 3rd.*

A B C D E F G A = A natural minor scale

[Try and measure the intervals between the notes of the A natural minor scale and see if they match the Minor Scale formula given above!]

The Relationship Between the Minor and Major Scales

As we have already discussed the concept of scale, how to make major and minor scales, it is time to examine the relationships between the major and minor scales. This will help us visualize the function of modes as well. Understanding how major and minor scales (or any scales in general) are related is necessary to comprehend the concept of **functional harmony**, which refers to the idea that each note plays different roles, or has a different function, in different positions of a scale.

Since we know that a *relative minor* has the same notes as its *"relative major,"* they must be related to each other in some way. Let us examine the formulae used to make the two scales

Major Scale Formula = **TTSTTTS**
Minor Scale Formula = **TSTTSTT**

Since we know that the relative minor scale begins at the 6th degree of the major scale, let's see how that affects the intervals once we start making a scale on the 6th degree

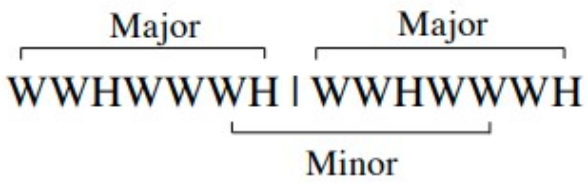
C	D	E	F	G	A	B	C					
					A	B	C	D	E	F	G	A
T	***T***	***S***	***T***	***T***	***T***	***S***	***T***	***T***	***S***	***T***	***T***	

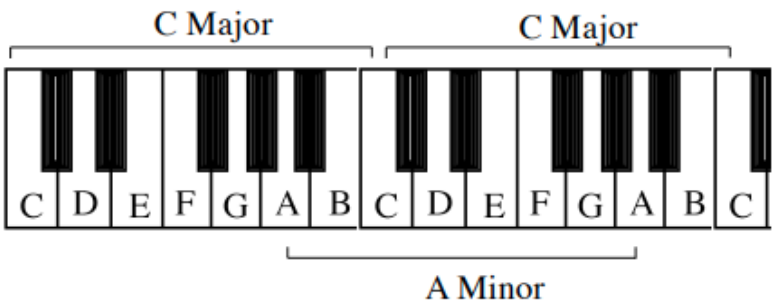
Looking at the minor scale this way, it is easy to notice that a relative minor scale can basically be viewed as a major scale beginning on the 6th degree. This causes the order of intervals and, therefore, the notes and diatonic chords (as we have previously discussed) to shift. The same concept applies for modes – if you start on the 2nd degree of the major scale, the interval formula would be TSTTTST – which makes the **Dorian mode**. The mode beginning on the 6th degree will have the same intervals as a Natural Minor scale and is called the **Aeolian mode.** We will discuss the relationships and usage of these modes and scales in a later section.

Ultimately, we need to understand how scales are functionally related, and how functions change when degrees change.

e.g. Even though the C note fulfils the function of the Tonic in the C major scale, it is the Minor 3rd in the A minor scale. This drastically changes our perception of the note within the scale. That is the basic idea of functional harmony.

The relationship between the Major and Minor scales can be depicted using the following diagrams:





Having understood the relationship between the major and minor scales, and having covered the concept of *degrees,* we can attach *qualities* to degrees to create intervals bigger than a semitone and tone. Qualities are separated into two categories – *Perfect and Imperfect.* Our *major* and *minor* qualities are classified as *Imperfect*, however they are rarely referred to as such. The Perfect quality is encountered at the 4th and 5th degrees.

For example, both the major and minor scales begin with a tone/whole step. That interval is also knows as a major 2nd. Then, if we attach a semitone to that it become a *minor 3rd* – **TS** is the beginning of a minor scale formula, therefore a minor scale has a minor 3rd. Adding another semitone will create **TT** which is a *major 3rd*, therefore a major scale has a major 3rd. These are the *qualities* of intervals/degrees. This will be covered in further detail within the section relating to chords. Down below are all the degrees with their respective qualities in the major and minor scales.

Major Scale	**Minor Scale**
Major Second	Major Second
Major Third	Minor Third
Perfect Fourth	Perfect Fourth
Perfect Fifth	Perfect Fifth
Major Sixth	Minor Sixth
Major Seventh	Minor Seventh

There are other ways to refer to the notes of a major scale and this other system will be given below.

Tonic
Supertonic
Mediant
Subdominant
Dominant
Submediant
Leading Tone

[In a natural minor scale the 7th note will be called the subtonic – a tone below the tonic.]

<u>Chords</u>

After making simple scales like major and minor, we can move on to finding *chords* within that scale. Each scale has a group of chords unique to the scale itself – this concept is known as **diatonic chords.** The simplest way to understand it is that these chords belong in the scale they are derived from. Let's try making some using the G major scale.

The simplest chord we can learn is known as a **Triad.** As the name suggests, it is made up of *three notes* (tri). By using *degrees* we can write down the formula for a triad.

Triad Formula = 1 3 5

[This is simply a blank slate for a simple triad. This doesn't tell us if it is major, minor, diminished, or augmented unless we specify the ***qualities of the degrees*** *we're using. Moreover, this is not the only triad formula we have as the 3rd can be replaced by the 2nd and 4th to make suspended triads/chords. e.g. 1 2 5, 1 4 5 Don't worry about that for now!]*

Let's make some chords in the <u>G major scale</u> by using 1 3 5 from each degree of the scale.

G A B C D E F# G

The first one is simple – <u>*G B D*</u> because those are the 1st, 3rd, and 5th notes of the G major scale. Then we can move to A *within the G major scale itself* and make another triad = <u>*A C E*</u>. You can continue like this until you reach the last note – F#. You should have *7 triads* in total.

The next step is being able to name the chords we've made. The first note of the chord/scale decides the name of the chord/scale. It is also known as the **root** or **tonic** of a chord/scale. So if it starts with F, it will be an F scale/chord unless specified otherwise. But how do we know if a chord is major, minor, diminished, or augmented? We do that by ***analysing the intervals.***

<u>*e.g.*</u> From the 1st note of G major scale (G) to the 3rd note (B) we have *two tones*. This is known as a **Major 3rd** interval. As the name suggests, major is bigger than minor, which itself means small. And by how much? A <u>semitone</u>. Using that logic, we can understand that a **Minor 3rd** would be a *semitone behind* the **Major Third**. Therefore:

Major 3rd = Tone + Tone
Minor 3rd = Tone + Semitone

When we understand how to identify major and minor thirds within triads, we can name the chord as major or minor according to the <u>*quality of the third*</u> it possesses.

But *what about the 5th?* The 5th is usually known as the **Perfect 5th.** This is because major and minor intervals are actually *imperfect intervals* – but it is rare to refer to them as such. All you need to remember is major and minor chords and scales have *Perfect 5th* and any change in the 5th will no longer retain its perfect quality. Measuring the distance from the 1st to the 5th note (G to D in G major) will show is that the Perfect 5th is 3 tones and 1 semitone.

e.g.	**Diminished 5th**	-	**Perfect 5th**	-	**Augmented 5th**
	(semitone behind perfect 5th)				*(semitone ahead of perfect 5th)*
	3 tones		3 tones + 1 semitone		4 tones

Now, if we **analyse the 3rds and 5ths** we come up with this progression of *diatonic chords*.

G major scale
G A B C D E F# G

Diatonic Chords
G B D = G major
A C E = A minor
B D F# = B minor
C E G= C major
D F# A = D major
E G B= E minor
F# A C = F# diminished

[Measuring the 3rds and the 5ths will show which ones have major and minor 3rds and that the 5th of the chord made on the 7th degree is a diminished 5th.]

Since *all major scales share the same formula*, we will find the <u>same order of chords within every major scale</u>. This means that we can translate them into Roman numerals to apply them to all major scales. *Uppercase numerals indicate Major chords, Lowercase numerals indicate Minor chords.* The empty dot above the chord made from the 7th degree indicates a *Diminished chord.*

Major Scale
7 diatonic chords
I ii iii IV V vi vii°
[This works for minor scales, modes, and other scales as well – that is a topic for future discussion]

Therefore, we write chord progressions like this:

I V vi IV *(Axis of Awesome chord progression)*
ii V I *(a common Jazz chord progression)*

This allows us to use any *key* or *scale* (synonymous in this context) if we remember the notes in that scale.

Another perspective on making chords is the concept of *stacking thirds*. Taking the example of F major scale, we can measure the distances between the notes to see the thirds hidden within the triads.

F major scale
F G A Bb C D E F

If we measure the distance between F and A we will find that it spans two tones. As previously discussed, that makes a Major Third. On the other hand, the distance between A and C is a tone and a semitone, which makes a Minor Third. Therefore, we find a **"stack" of thirds** – a Major Third with a Minor Third on top. In the minor chords, we notice that the opposite is true – it is made up of a Minor Third and a Major Third. A diminished chord will be made of two Minor Third intervals. This knowledge will help you further when you learn about seventh chords, because that requires simply adding another Third to the stack!

F A C = F major [Major Third + Minor Third]
G Bb D = G minor [Minor Third + Major Third]
A C E = A minor [Minor Third + Major Third]
Bb D F = Bb major [Major Third + Minor Third]

C E G = C major [Major Third + Minor Third]
D F A = D minor [Minor Third + Major Third]
E G Bb = E diminished [Minor Third + Minor Third]

The Circle of Fifths

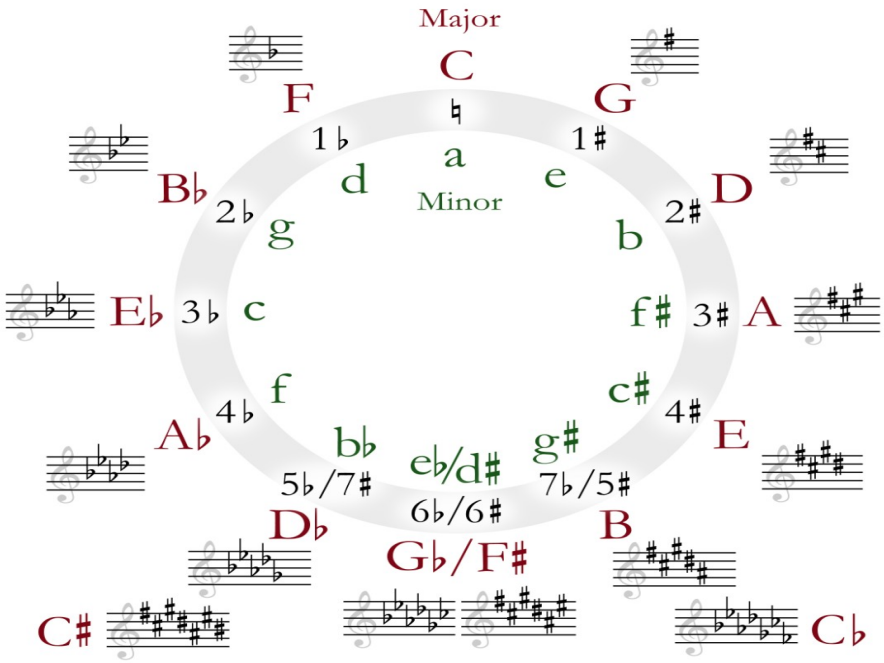


Fig. 1

The **Circle of Fifths** is an extremely useful tool to help us memorize scales as well as remember the diatonic (and non-diatonic) relationships between scales and chords. It is known as the Circle of Fifths because when going *clock-wise*, we go up in 5^{ths}. It can be useful to visualize it like a clock with C at the 12th position.

e.g. In the C major scale (C D E F G A B C) the note on the 5th degree (G) is going to be the next note clockwise in the Circle of Fifths. Hence, we are *moving up from the 1st to the 5th degree*; moving up in 5^{ths}.

Conversely, when we go *anti-clockwise*, we go in Fourths. So we when we read the Circle anti-clockwise, we call it the **Circle of Fourths.** Why is that? It is because of the concept of *inversions*. Let's look at an example.

e.g. In the C major scale, moving from C to G takes through 5 notes (including C and G). However, moving from G to C takes us through 4 notes (including G and C). This is called an **inversion**. Therefore, a perfect 5th is an inversion of a perfect 4th and vice versa. Try it out with other scales! C to F, for example, is a 4th, F to C is a 5th and so on.

As mentioned, visualizing the Circle of Fifths as a clock helps us keep track of how many sharps and flats each scale has. This *helps us make scales mentally much faster* than TTSTTTS (major scale formula – but only if we remember the order of the notes!

We have talked a lot about C major and by now we have internalized the fact that C major scale has all natural notes (no sharps or flats). Therefore, when going *clock-wise* in the Circle of Fifths we add **sharps**, and when we go *anti-clockwise* we add **flats**. Therefore, if C has zero sharps, going to G will mean it has 1 sharp. Conversely, if C has zero flats, F major will have one flat.

Remembering where we add sharps and flats is important.
Going clockwise = Add new **sharp at 7th degree**
Going anti-clockwise = Add new **flat at 4th degree.**

Let's look at some examples. Use the diagram on the first page to confirm that we are going in fifths and fourths.

Major Scales Up in 5ths
C D E F G A B C
G A B C D E F# G = 1 sharp *(added at 7th degree)*
D E F# G A B C# D = 2 sharps *(new sharp added at 7th degree)*
A B C# D E F# G# A = 3 sharps
(etc.)

Major Scales Up in 4ths
C D E F G A B C
F G A Bb C D E F = 1 flat *(added at 4th degree)*
Bb C D Eb F G A Bb = 2 flats *(new flat added at 4th degree)*
Eb F G Ab Bb C D Eb = *3 flats*
(etc.)

Note how each successive scale retains the sharp/flat added in the previous scale.

***Exercise**: Make major scales clock-wise and anti-clockwise from C major until you have all 7 notes sharp or flat.*

In a previous section, we covered what a *Relative Minor* is and where it is found within the major scale. The inner circle in Fig. 1 also gives us the Relative Minors for each scale. Look at C – the lowercase 'a' refers to A minor, thus specifying that C major's Relative Minor is A (natural) minor. For G major, the Relative Minor is E (natural) minor.

P.S.- Natural minors are the common minor scales made with TSTTSTT formula, as compared to Harmonic and Melodic Minor scales which we will get to later.

(This site has an interactive Circle of Fifths with the *Diatonic Chords* for the chosen scale specified within the inner circle.)

The Fig. 2 given below shows that the *Tonic* (root note, first note of the scale) selected is C, therefore the notes within outer white boxes are all notes of the C major scale. You can see that only the natural notes are within the white boxes, thus confirming that it is the C major scale.

In the inner circle, we have uppercase and lowercase lower numerals specifying the *Diatonic Chords in a Major Scale*. A previous section of initial theory covers the diatonic chords of a major scale. Therefore, *chords made on the 1st, 4th, 5th degrees are Major, ones made on the 2nd, 3rd, 6th degrees are Minor, and the chord made on the 7th degree is diminished.* Below is the order in Roman numerals, as discussed previously.

I ii iii IV V vi vii°

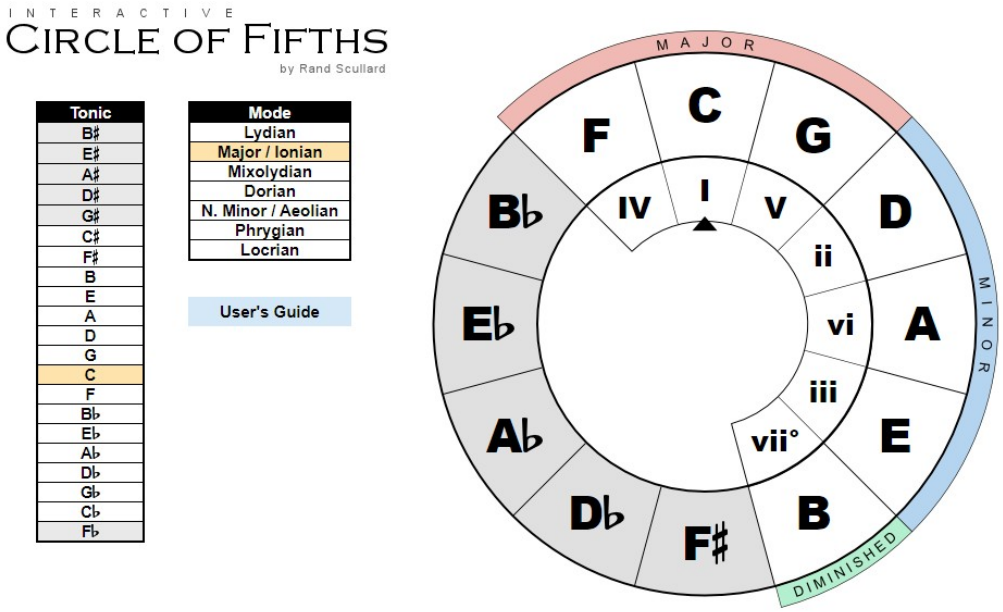


Fig. 2

Knowing the order of the diatonic chords in a major (and minor scale) will help us easily recall which chords are major, minor, or diminished whichever scale we are using. All we need to remember are the degrees and which degree makes a major, minor, or diminished chord.

For the minor scale, we will have the same chords but in a different order (since the relative minor has the same notes as its 'relative major' scale, only ordered differently). Therefore the 6th note and chord of a Major scale becomes the 1st note of the Minor scale. The Fig. 3 below will show the diatonic chords within the A Minor scale, which is the Relative Minor of C major scale.. Note how we have the same chords, except at different degrees of the scale.

Therefore, the order of the Diatonic Chords in a Natural Minor scale is:

i ii° III iv v VI VII

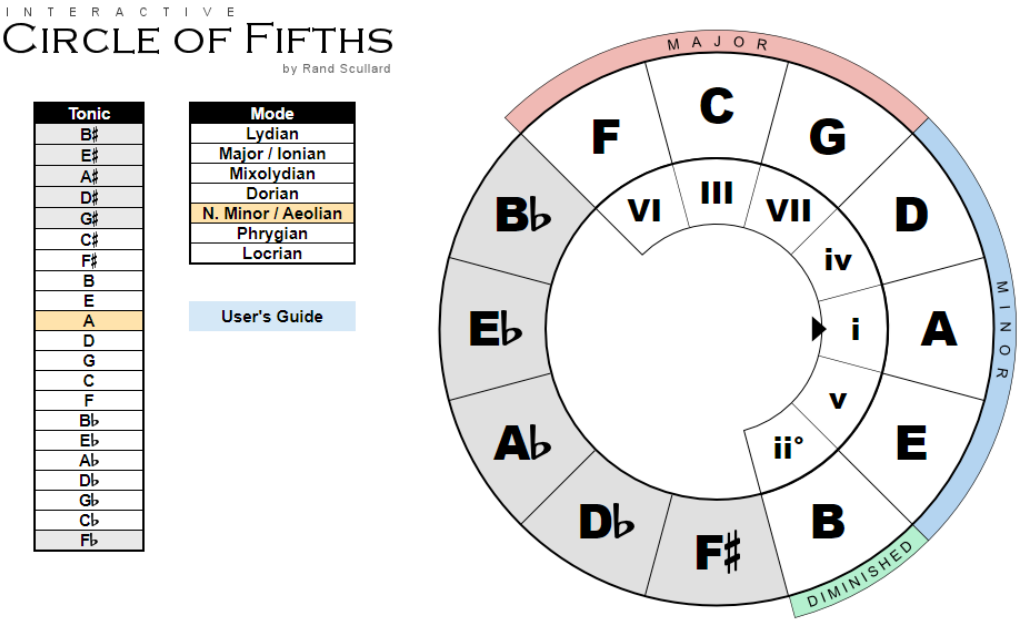


Fig. 3

Apart from helping us remember how many sharps and flats we have in a scale (if we remember the order F C G D A E B), the Circle of Fifths also assists us in understanding **functional harmony.**

The basic understanding of functional harmony is that each note or chord, when at a specific degree, has one function whereas the same note at another degree of a scale or chord will play a different function. We will get to this concept later, but it is important to remember that different notes play different functions when ordered at different degrees.

So far, we have covered the **usefulness of the Circle of Fifths** with these points:

1. It helps us *identify scales* without the formula if we visualize the circle and remember which major scales have how many sharps/flats.

2. Knowing a major scale automatically brings us to the relative minor – therefore allowing us to *make natural minor scales* quicker. This works with modes as well, as we will discuss further.

3. It helps us *keep track of the diatonic chords in a major or minor scale.* The major chords are always together, and so are the minor chords. The diminished chord lies at the end of the scale in the order of Fifths (clockwise).

4. A topic that will be covered in the next section will highlight how the Circle of Fifths can help us make *non-diatonic chord progressions/lines* based on the relationships of the mode/scale in question. This works by borrowing chords. A simple example will be given below.

e.g. The song Gulabi Aankhein or Hit The Road Jack both have an E7 chord even though the key (scale) of the song is in A minor. By the order of diatonic chords, the 5th chord (which in this case is made from the note E in the minor scale) should be a minor chord. However, the chord is a Major chord, which means that it has been borrowed from the ***parallel major*** *– the A major scale! Parallel scales are merely scales that begin on the same tonic. Like A major is a parallel major key to A minor and so on. Such a method of borrowing chords is very common. The previously mentioned chord progression in Gulabi Aankhein and Hit The Road Jack is Am G F E7 or i VII VI V in the A minor key. This is also known as the Andalusian cadence.*

Exercise – *Another example of a the same progression in a different key is Neele Neele Ambar Par or Sultans of Swing. The key of the song is D minor – try to find out the chord progression using the diatonic chords with borrowing from the parallel key.*

Scales Continued

Apart from the general major and minor scales, we have additional scales such as the *modes* of the major scale and the *Harmonic* and *Melodic* minor scales. Sometimes, along with thinking of them as separate scales, perceiving them as alterations of the major and minor scales helps us understand the function of the notes we're playing. This will become clearer as we discuss the Harmonic and Melodic minor scales below.

Usually, music theory textbooks treat the Harmonic and Melodic minor scales as separate scales. The Harmonic Minor scale is essentially the Natural Minor scale with a raised 7th – the leading tone.

A Natural Minor

A B C D E F G A

A Harmonic Minor

A B C D E F G# A

However, looking at the Harmonic minor scale as a Natural minor with a raised 7th is a more appropriate approach. This is because composers whose works were analysed to provide theoretical examples were not thinking "I am using the Harmonic minor scale," but were rather thinking "I am going to use the leading tone – a raised 7th from the parallel major key for a stronger pull back towards the Tonic."

In a major scale, the major 7th is a semitone away from the Tonic. This creates a tension that resolves back at the Tonic note. This is why a raised 7th is known as a leading tone – it leads back to the Tonic. A Natural minor scale, on the other hand, has a Minor 7th which is a tone away from the Tonic. Therefore, "borrowing" the Major 7th creates that strong pull in a Minor scale.

In the same vein, a Melodic Minor has a raised 6th – a Major 6th. But only when the scale is played in the ascending order. Why? This is a quirk of composition due to which it is better to view it as a Natural Minor scale with a raised 6th and 7th than a separate scale that has to be played differently in the ascending and descending orders.

A Natural Minor

A B C D E F G A

A Melodic Minor

A B C D E F# G# A *(ascending)*

A G# F E D C B A *(descending)*

Now, we will talk about the modes of a major scale. We will be considering each mode as an alteration of the major scale as well, to help us memorize them and the function of each alteration. The major scale itself is a mode – the Ionian modes. You will come to see that the names are slightly complicated, this is due to the fact that they are the names of Greek tribes. This will make memorizing them slightly complicated, but there will be a mnemonic mentioned to help!

If the major scale is an Ionian mode, playing the major scale with its 2nd as the Tonic will create the Dorian mode. Let us use F major as an example. We will note the degrees in numbers below, each unaltered number corresponds to a major scale. Therefore 1 2 3 4 5 6 7 will be the degrees of whatever major scale we use. This will help us see how a mode compares to the major scale.

F Ionian mode (Major Scale)

F G A Bb C D E F

1 2 3 4 5 6 7

G Dorian mode

G A Bb C D E F G

1 2 b3 4 5 6 7

We can do this for each degree of the Major scale to find the modes, as we have done below. Remember that the Relative Minor is found at the 6th degree of the Major Scale. It will also be a mode of the Major scale – the Aeolian mode.

F Ionian mode (Major Scale)

F G A Bb C D E F

1 2 3 4 5 6 7

G Dorian mode

G A Bb C D E F G

1 2 b3 4 5 6 b7

A Phrygian mode

A Bb C D E F G A

1 b2 b3 4 5 b6 b7

Bb Lydian mode

Bb C D E F G A Bb

1 2 3 #4 5 6 7

C Mixolydian mode

C D E F G A Bb C

1 2 3 4 5 6 b7

D Aeolian (Natural minor scale)

D E F G A Bb C D

1 2 b3 4 5 b6 b7

E Locrian mode

E F G A Bb C D E

1 b2 b3 4 b5 b6 b7

We will talk about how to hear intervals, with relevant examples, further in our discussion. For now, we will continue with scales like the *Pentatonic*. As the name suggests, “penta” refers to five, therefore the scale will have five notes, instead of 7 like the major and minor scales.

The major pentatonic scale has the following notes of the major scale – 1 2 3 5 6. Therefore, a C major pentatonic scale will be – C D E G A.

The minor pentatonic scale has the following notes of the minor scale – 1 3 4 5 7. Therefore, an A minor pentatonic scale will be – A C D E G.

As you can see, the major pentatonic scale will share the same notes as the pentatonic scale made from its relative minor. Another notable “scale” is the blues scale, which has many variations. However, when referring to a six-note scale, the blues scale is a minor pentatonic scale with a b5 – the “blue note.” Theoretically, we perceive it as an alteration of a minor pentatonic, similar to how we talked about the harmonic and melodic minor scales.

Having discussed modes, lets move further to analyse the *modes* of the *harmonic* and *melodic* minor scales. This may seem like an advanced concept – and it is – but understanding how notes are changed from one scale to another is an important skill to have as a musician. Whatever your instrument may be, try playing these new scales over and over to listen to how they sound. Your musical vocabulary and relative pitch – the ability to identify intervals in relation to one another – will improve.

Since the harmonic minor scale has the leading tone, a major 7th, the mode changes will be as follows.

<u>A Harmonic minor scale</u>

A B C D E F G# A

1 2 b3 4 5 b6 7

<u>Locrian 6/ Locrian 13</u>

B C D E F G# A B

1 b2 b3 4 b5 6 b7

<u>Ionian #5</u>

C D E F G# A B C

1 2 3 4 #5 6 7

<u>Dorian #4/ Dorian #11</u>

D E F G# A B C D

1 2 b3 #4 5 6 b7

<u>Phrygian Dominant</u>

E F G# A B C D E

1 b2 3 4 5 b6 b7

<u>Lydian #2</u>

F G# A B C D E F

1 #2 3 #4 5 6 7

Super Locrian bb7

G# A B C D E F G#

1 b2 b3 b4 b5 b6 bb7

Likewise, the modes of the Melodic minor scale will be given below.

A Melodic minor

A B C D E F# G# A

1 2 b3 4 5 6 7

Dorian b2

B C D E F# G# A B

1 b2 b3 4 5 6 b7

Lydian Augmented

C D E F# G# A B C

1 2 3 #4 #5 6 7

Lydian Dominant

D E F# G# A B C D

1 2 3 #4 5 6 b7

Mixolydian b6

E F# G# A B C D E

1 2 3 4 5 b6 b7

Aeolian b5/Locrian #2

F# G# A B C D E F#

1 2 b3 4 b5 b6 b7

Super Locrian/Altered Scale

G# A B C D E F# G#

1 b2 b3 b4 b5 b6 b7

Chords Continued

Having examined the various omissions and alterations of notes within scales, we can learn the chords of the harmonic and melodic minor scales, as well as diatonic seventh chords. As previously discussed, these are the diatonic chords of a major and minor scale.

Major Scale

I ii iii IV V vi vii°

Minor Scale

i ii° III iv v VI VII

In the harmonic minor scale, raising the seventh degree of the affects the diatonic chords of the natural minor scale. Specifically, the 3rd, 5th, and 7th triads are affected. Lets use A harmonic minor as an example to show the diatonic triads of a harmonic minor scale.

A Harmonic minor
A B C D E F G# A

Diatonic Triads
A C E = A minor = i
B D F = B diminished = ii°
C E G# = C augmented = III+
D F A = D minor = iv
E G# B = E major = V
F A C = F major = VI
G# B D = G diminished = vii°

Likewise, raising the 6th degree is going affect other triads when we consider the melodic minor scale. Specifically, the 2nd, 4th, and 6th triads will change. Using the A melodic minor scale as an example, we come up with the following chords.

A Melodic minor
A B C D E F# G# A

Diatonic Triads
A C E = A minor = i
B D F# = B minor = ii
C E G# = C augmented = III+
D F# A = D major = IV
E G# B = E major = V
F# A C = F# diminished = vi°
G# B D = G diminished = vii°

When making a 7th chord, we add another third to the stack of thirds that is a simple major, minor, or diminished triad. We can have the following 7th chords. Examples will be given from the C major/minor scale.

Major 7th = 1 3 5 7 = Major 3rd + Minor 3rd + Major 3rd *(C E G B)*

Minor 7th = 1 b3 5 b7 = Minor 3rd + Major 3rd + Minor 3rd *(C Eb G Bb)*
Dominant 7th = 1 3 5 b7 = Major 3rd + Minor 3rd + Minor 3rd *(C E G Bb)*
Minor-Major 7th = 1 b3 5 7 = Minor 3rd + Major 3rd + Major 3rd *(C Eb G B)*
Half Diminished = 1 b3 b5 b7 = Minor 3rd + Minor 3rd + Major 3rd (C Eb Gb Bb)
Fully Diminished = 1 b3 b5 bb7 = Minor 3rd + Minor 3rd + Minor 3rd (C Eb Gb Bbb)
Diminished Major 7th = 1 b3 b5 7 = Minor 3rd + Minor 3rd + Augmented 3rd *(C Eb Gb B)**
Augmented 7th = 1 3 #5 b7 = Major 3rd + Major 3rd + Diminished 3rd *(C E G# Bb)*
Augmented Major 7th = 1 3 #5 7 = Major 3rd + Major 3rd + Minor 3rd *(C E G# B)*

**rare*

The order of diatonic seventh chords in a major and minor scales are given below.

Major Scale

Imaj7 iim7 iiim7 IVmaj7 V7 vim7 viiø

Minor Scale

im7 iiø IIImaj7 ivm7 vm7 VImaj7 VII7

Harmonic Minor scale

iMaj7 iiø III+Maj7 ivm7 V7 VImaj7 vii°

Melodic Minor scale

iMaj7 iim7 III+Maj7 IV7 V7 viø viiø

[maj7 refers to a major 7th chord, m7 refers to a minor 7th chord, and simply a 7 denotes a dominant 7th chord. The ø symbol means half diminished, which is a diminished chord with a minor seventh. The 7th chord of the harmonic minor scale will be a fully diminished chord, with a diminished 7th.]

Reading Sheet Music

An important skill for most musicians is the ability to read sheet music. So far, we have covered musical concepts without the usage of sheet music – this shows that it is possible to have an in-depth understanding of theory without knowing how to read sheet music. However, knowing the basics will enhance your musical vocabulary.

There are two crucial aspects to reading sheet music. The first is rhythm, or time, and the second is melody. As Fig 1. shows, sheet music is written on five lines – known as the *staff*. Whether a note comes on a line or a space will define what sound will be played, and the notation of the note will tell us how long to play it for. Likewise, music consists of silence between notes which are signified by rests. Let us first look at the notations available to signify rhythm values and rests in sheet music.

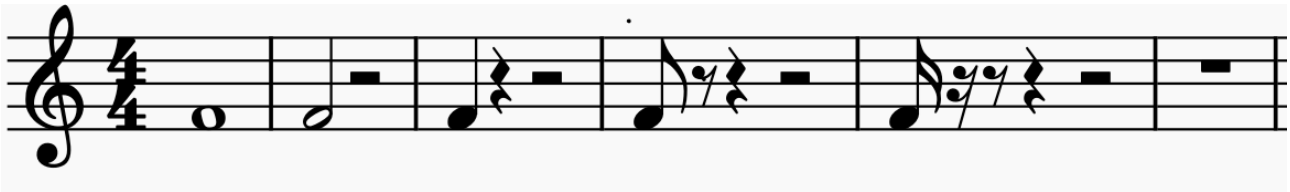
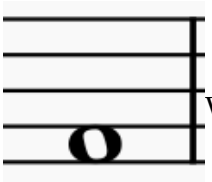
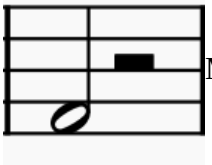


Fig 1.

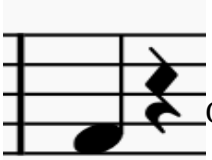


Whole Note/Semibreve = 4 beats

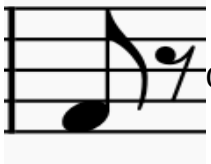
Whole Note Rest = 4 beats



Minim/Minim Rest = 2 beats



Crochet/Crochet Rest = 1 beat



Quaver/Quaver Rest = ½ beat